

2 People who become infected with human immunodeficiency virus (HIV) are at risk of developing HIV/AIDs, particularly if antiretroviral therapy (ART) is not available.

(a) In people infected with HIV, the use of ART also helps to reduce transmission of the virus to uninfected people.

Outline **two** control methods, other than ART, that can be used to reduce the transmission of HIV.

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..... [2]

In people with HIV/AIDs, a serious lung disease known as pneumocystis pneumonia can result from infection by an opportunistic pathogen known as *Pneumocystis jirovecii*.

Fig. 2.1 shows *P. jirovecii* cells in one stage of their life cycle, as seen using a light microscope at a magnification of  $\times 600$ .

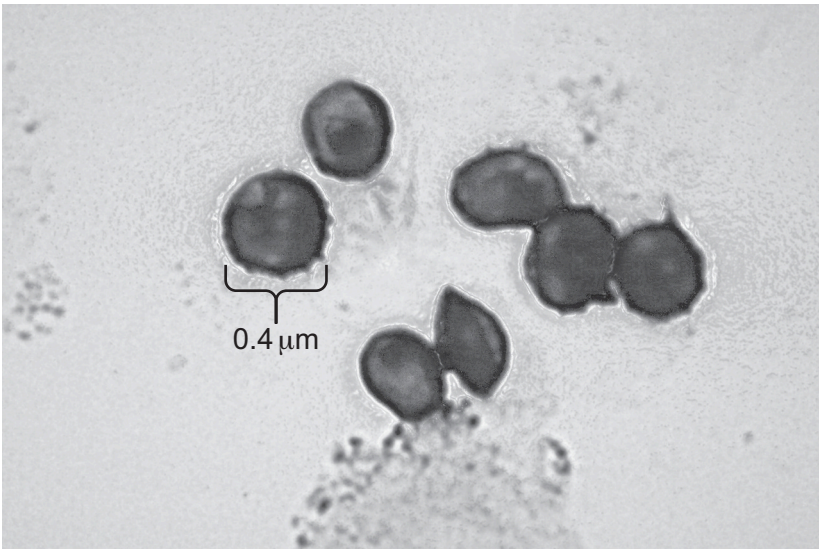


Fig. 2.1

(b) Define magnification.

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..... [1]

- (c) Fig. 2.1 shows that *P. jirovecii* is a unicellular organism. Although the cells of many species of bacteria are the same size as those of *P. jirovecii*, research concluded that the organism is a eukaryote and is **not** a bacterium.

In 1988, analysis of ribosomal RNA (rRNA) resulted in *P. jirovecii* being classified as a fungus.

- (i) Studies of the structure of *P. jirovecii* have identified that the cell wall is made of polysaccharides such as chitin and 1,3-β-D-glucan.

Explain why this feature helped scientists to confirm that *P. jirovecii* is **not** a bacterium.

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..... [1]

- (ii) Scientists have identified other features of the cell structure of *P. jirovecii*. Some of these are listed in Table 2.1.

Complete each row of Table 2.1 so that the table shows:

- four structural features identified in *P. jirovecii*
- one function for each structural feature
- whether the structural feature is present (✓) or absent (✗) in bacterial cells.

Table 2.1

| structural feature of <i>P. jirovecii</i> | function                            | present (✓) or absent (✗) in bacterial cells |
|-------------------------------------------|-------------------------------------|----------------------------------------------|
| ribosomes                                 | protein synthesis                   |                                              |
| smooth endoplasmic reticulum              |                                     |                                              |
| Golgi body                                | modification of proteins and lipids |                                              |
|                                           | aerobic respiration                 |                                              |

[3]

- (d) *P. jirovecii* can adhere (attach) to squamous epithelial cells of the alveoli and to the network of fibrous proteins that support the alveolar wall, known as the extracellular matrix (ECM). Examples of proteins in the ECM are elastin and collagen.

Adhesion (attachment) of *P. jirovecii* to alveolar epithelial cells and the ECM stimulates the growth of its population.

- (i) Cell surface glycoproteins known as gpA glycoproteins are essential in allowing *P. jirovecii* cells to adhere to alveolar epithelial cells and ECM proteins.

Suggest how a gpA glycoprotein is able to adhere to alveolar epithelial cells and ECM proteins.

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..... [2]

- (ii) One consequence of the pneumonia that results from *P. jirovecii* infection is a decrease in the quantity of oxygen that is delivered to body tissues.

Explain why a severe *P. jirovecii* infection results in a decrease in the quantity of oxygen that is delivered to body tissues.

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(e) *P. jirovecii* produces an enzyme known as 1,3-β-D-glucan synthase. The enzyme catalyses the synthesis of 1,3-β-D-glucan.

With reference to the mechanism of action of caspofungin, explain how the drug may be useful to treat cases of pneumonia caused by *P. jirovecii*.

[5]

[Total: 17]